

Windows Server 2025 Home Lab

LinkedIn Project Evidence - Active Directory, SMB, NTFS Permissions & Group Policy

Project objective: Simulate a common enterprise IT workflow by providing a domain user with secure access to a departmental file share and automatically deploying that share as a mapped drive through Group Policy.

Environment: Hyper-V | Windows Server 2025 DC01 | Windows 11 Enterprise C01 | jjadamslab.local | GG-IT-Users | jsmith

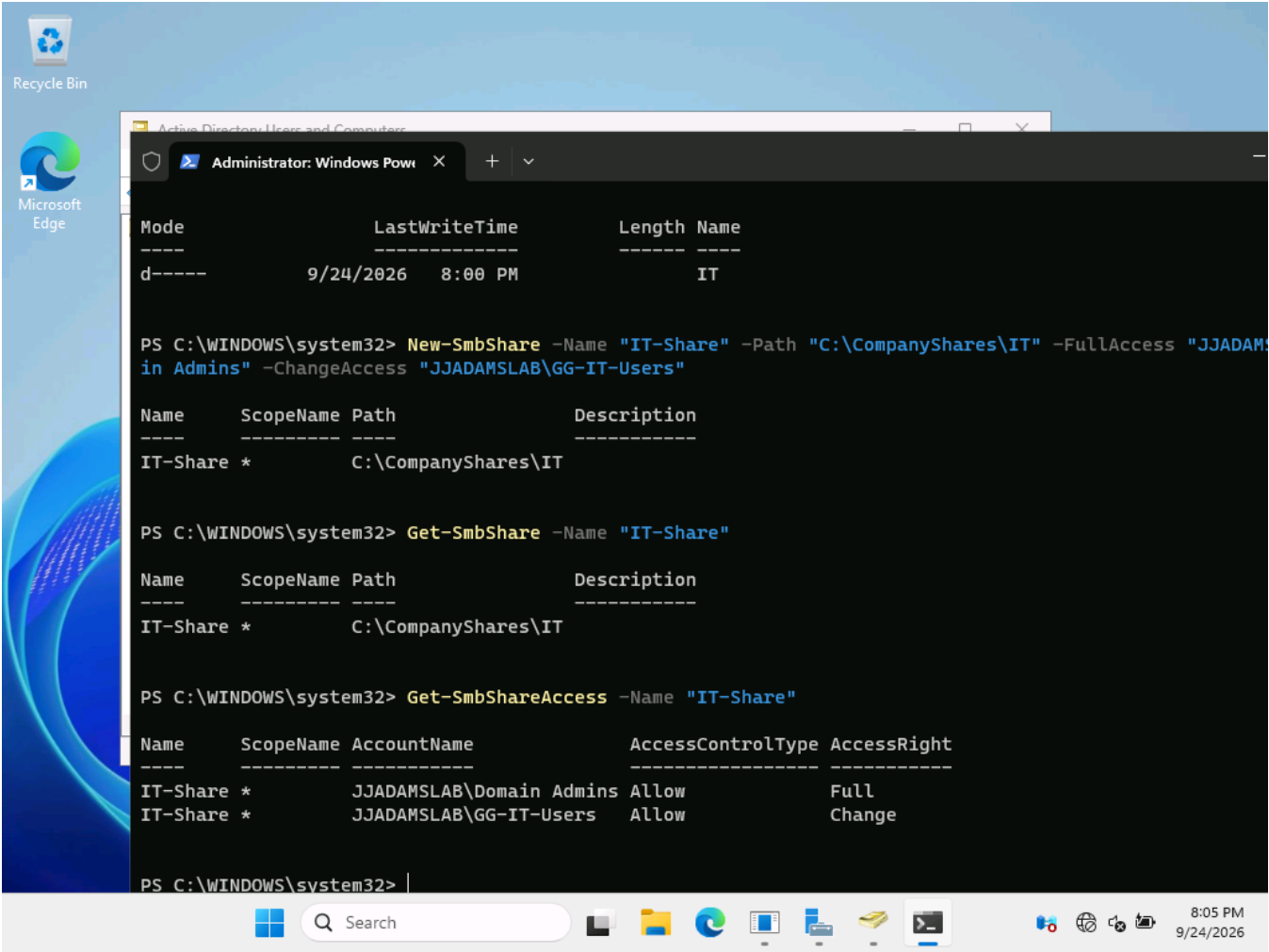
Project flow

1. Deploy SMB share
2. Configure NTFS permissions
3. Test domain-user access
4. Deploy mapped-drive GPO
5. Verify automatic mapping
6. Validate identity, mapping and file access

Final result: JJADAMSLAB\jsmith -> I: -> \\DC01\IT-Share -> Status OK

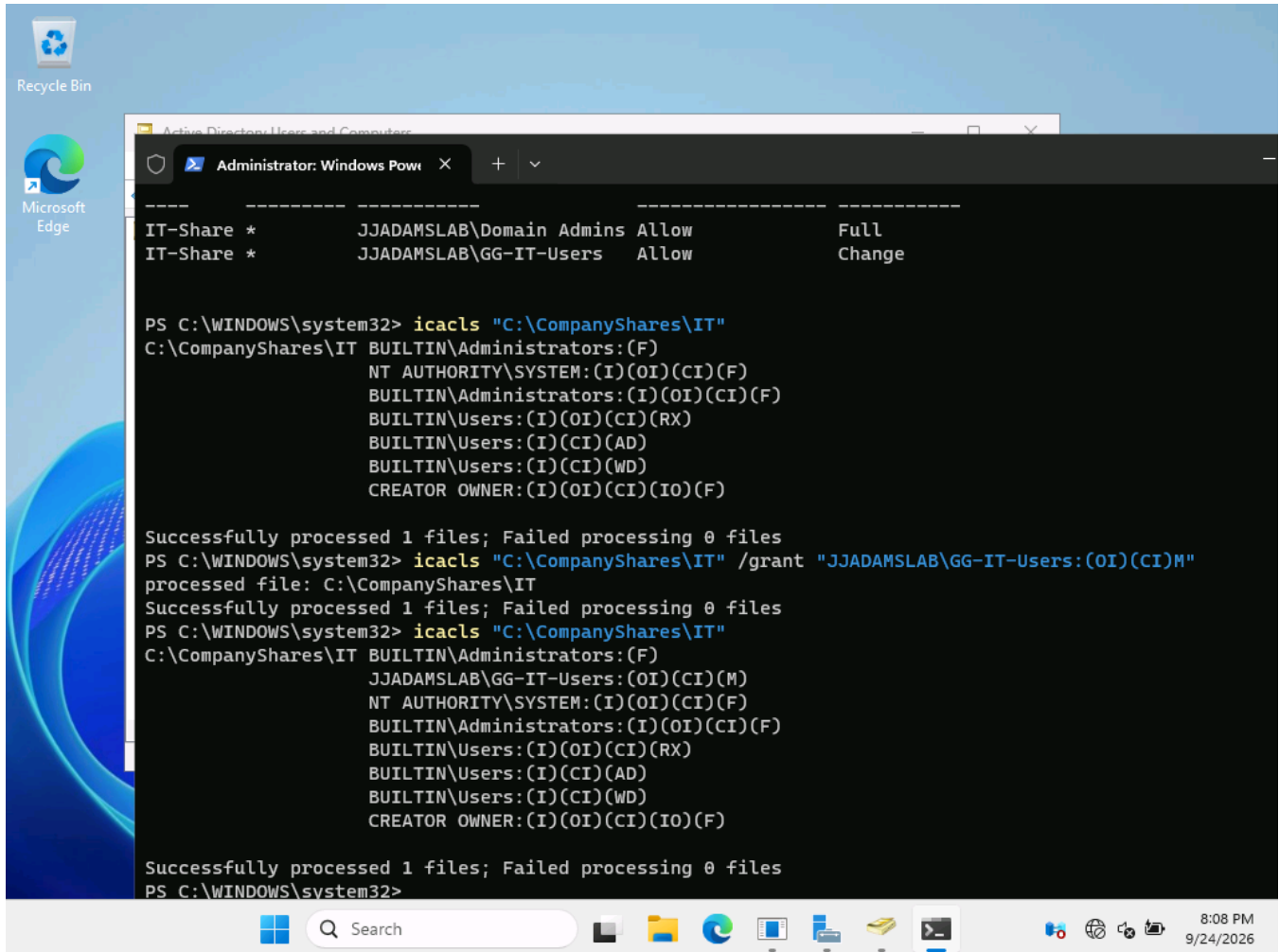
Step 1 - Deploy the IT SMB Share

Deployed the IT departmental share on DC01. This provides the central network location that IT users will access.



Step 2 - Configure NTFS Permissions

Assigned Modify access to the Active Directory security group GG-IT-Users, allowing group-based access instead of assigning permissions directly to individual users.



```
-----  
IT-Share *      JJADAMSLAB\Domain Admins Allow      Full  
IT-Share *      JJADAMSLAB\GG-IT-Users Allow      Change  
  
PS C:\WINDOWS\system32> icacls "C:\CompanyShares\IT"  
C:\CompanyShares\IT BUILTIN\Administrators:(F)  
NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)  
BUILTIN\Administrators:(I)(OI)(CI)(F)  
BUILTIN\Users:(I)(OI)(CI)(RX)  
BUILTIN\Users:(I)(CI)(AD)  
BUILTIN\Users:(I)(CI)(WD)  
CREATOR OWNER:(I)(OI)(CI)(IO)(F)  
  
Successfully processed 1 files; Failed processing 0 files  
PS C:\WINDOWS\system32> icacls "C:\CompanyShares\IT" /grant "JJADAMSLAB\GG-IT-Users:(OI)(CI)M"  
processed file: C:\CompanyShares\IT  
Successfully processed 1 files; Failed processing 0 files  
PS C:\WINDOWS\system32> icacls "C:\CompanyShares\IT"  
C:\CompanyShares\IT BUILTIN\Administrators:(F)  
JJADAMSLAB\GG-IT-Users:(OI)(CI)(M)  
NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)  
BUILTIN\Administrators:(I)(OI)(CI)(F)  
BUILTIN\Users:(I)(OI)(CI)(RX)  
BUILTIN\Users:(I)(CI)(AD)  
BUILTIN\Users:(I)(CI)(WD)  
CREATOR OWNER:(I)(OI)(CI)(IO)(F)  
  
Successfully processed 1 files; Failed processing 0 files  
PS C:\WINDOWS\system32>
```

Step 3 - Test Domain User Access

Validated SMB access from C01 using the domain user jsmith and created a test file to prove write access to the departmental share.

```
Windows PowerShell
Name                                     Type      TTL      Section  IPAddress
----
dc01.jjadamslab.local                  A          3600    Answer   10.20.30.10

PS C:\WINDOWS\system32> nltest /dsgetdc:jjadamslab.local
DC: \\DC01.jjadamslab.local
Address: \\10.20.30.10
Dom Guid: 2079fe5a-1bd6-40be-b336-cebdcbe09457
Dom Name: jjadamslab.local
Forest Name: jjadamslab.local
Dc Site Name: Default-First-Site-Name
Our Site Name: Default-First-Site-Name
Flags: PDC GC DS LDAP KDC WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET WS DS_8 DS_9 DS_10 KEYLIST DS_13
The command completed successfully
PS C:\WINDOWS\system32> net use \\DC01\IT-Share /user:JJADAMSLAB\jsmith *
Type the password for \\DC01\IT-Share:
The command completed successfully.

PS C:\WINDOWS\system32> dir \\DC01\IT-Share
PS C:\WINDOWS\system32> dir \\DC01\IT-Share
PS C:\WINDOWS\system32> echo Project 2 permission test > \\DC01\IT-Share\jsmith-test.txt
PS C:\WINDOWS\system32> dir \\DC01\IT-Share

Directory: \\DC01\IT-Share

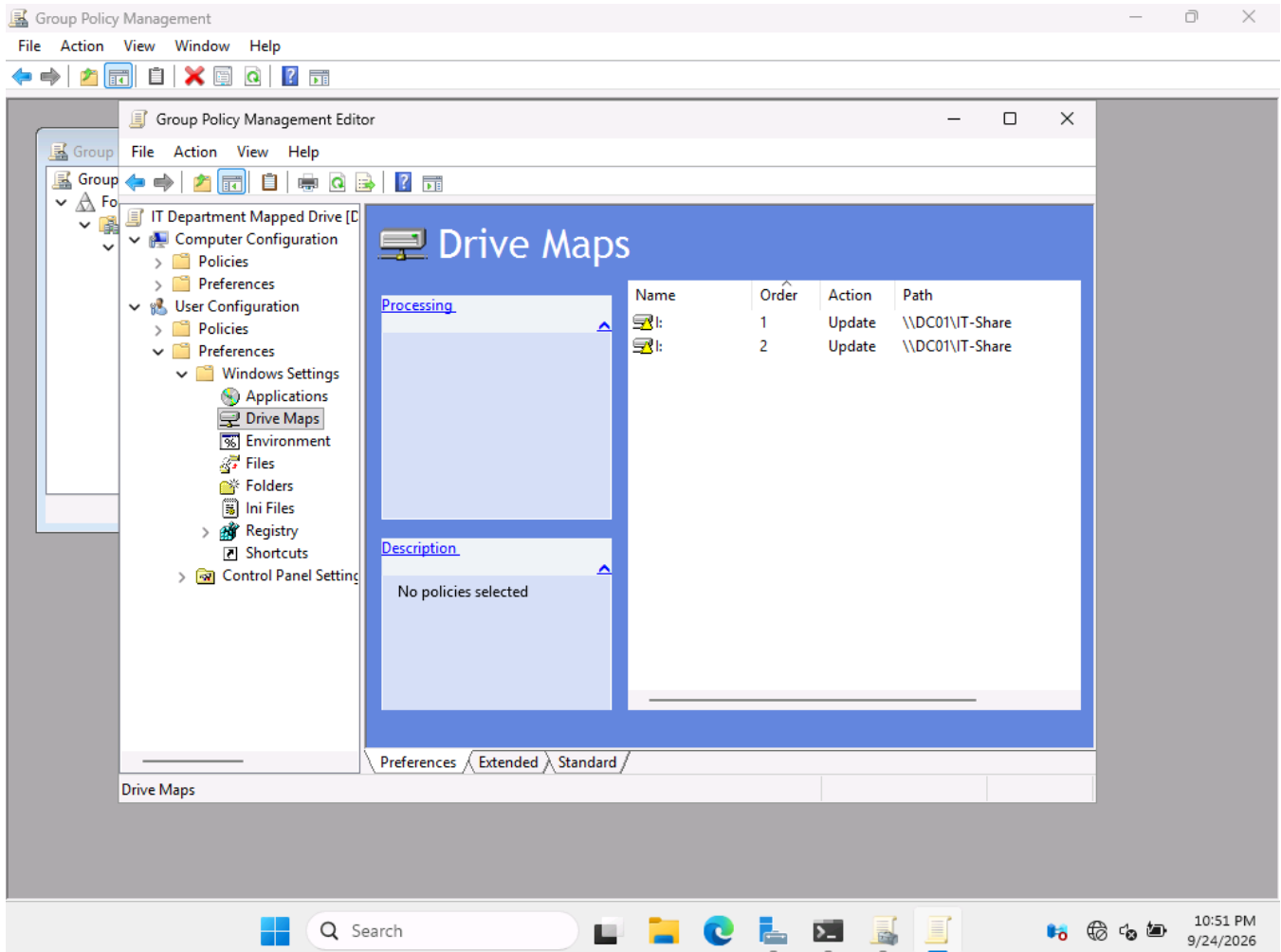
Mode                LastWriteTime         Length Name
----                -
-a-----          9/24/2026   8:45 PM             62 jsmith-test.txt

PS C:\WINDOWS\system32> |
```

Troubleshooting note: During testing, C01 was found outside the domain. After rejoining it to jjadamslab.local, domain authentication and user GPO processing were restored.

Step 4 - Deploy the GPO Mapped Drive

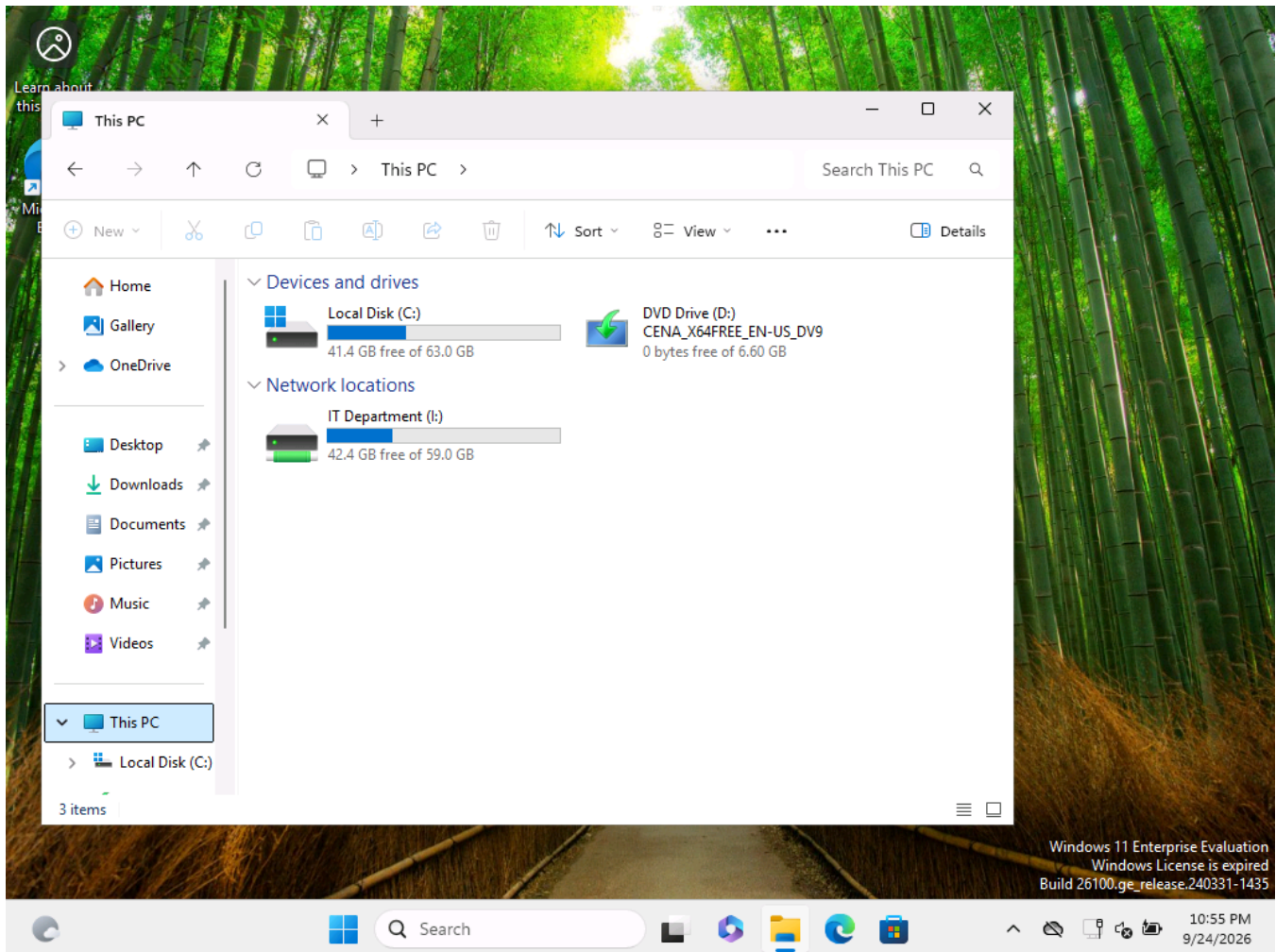
Configured Group Policy Preferences to map \\DC01\IT-Share as the IT Department I: drive for users in the IT OU.



Troubleshooting note: Manual SMB mapping worked, which helped isolate the remaining problem to Group Policy Preferences. A duplicate I: Drive Maps entry was removed before the final successful deployment.

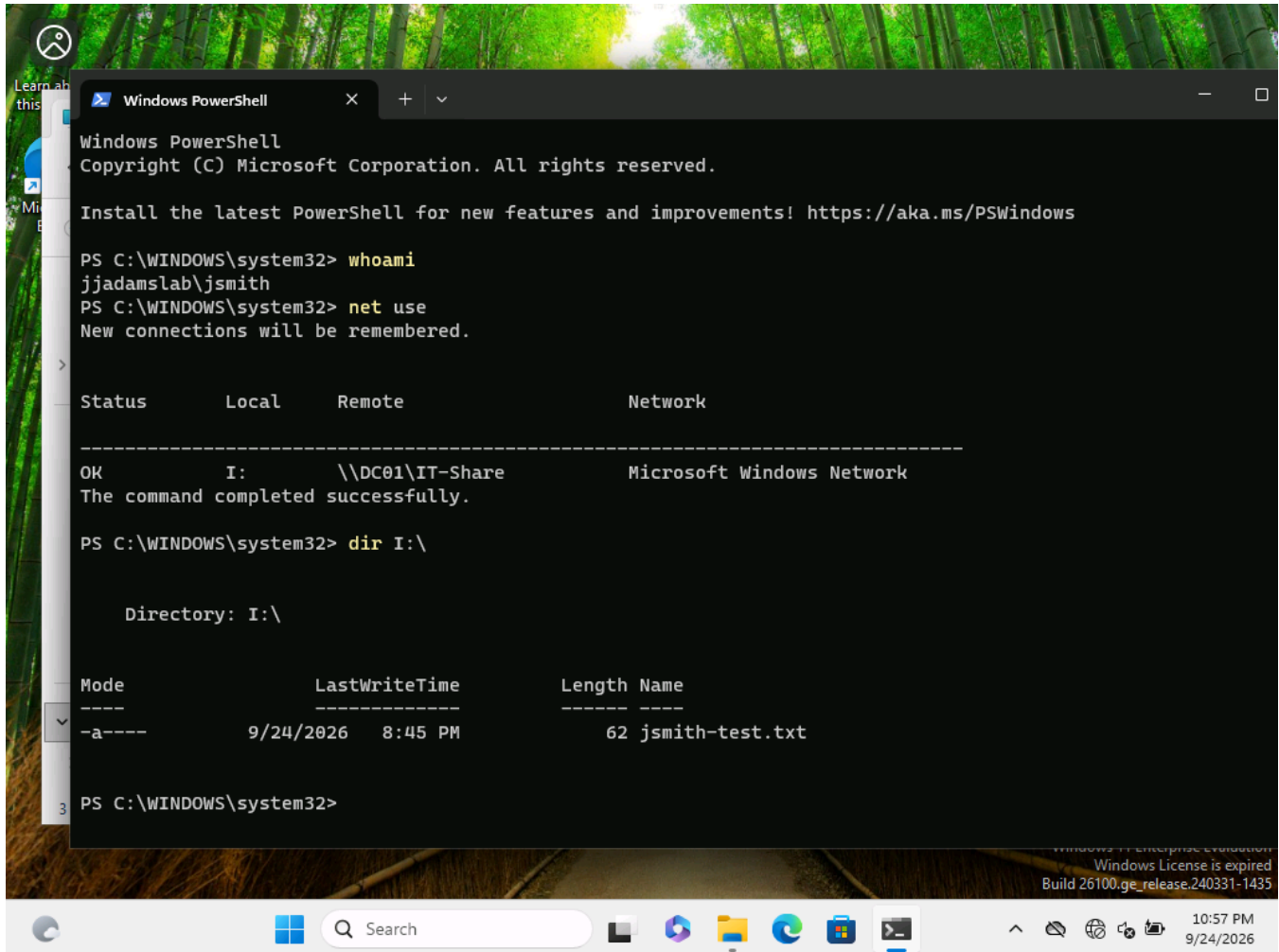
Step 5 - Verify Automatic Drive Deployment

After Group Policy refresh and domain sign-in, the IT Department (I:) drive appeared automatically in File Explorer on C01.



Step 6 - Final End-to-End Verification

Confirmed jjadamslab\jsmith, I: mapped to \\DC01\IT-Share with Status OK, and verified access to jsmith-test.txt.



The screenshot shows a Windows PowerShell terminal window with the following content:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> whoami
jjadamslab\jsmith
PS C:\WINDOWS\system32> net use
New connections will be remembered.
```

Status	Local	Remote	Network
OK	I:	\\DC01\IT-Share	Microsoft Windows Network

The command completed successfully.

```
PS C:\WINDOWS\system32> dir I:\
```

Directory: I:\

Mode	LastWriteTime	Length	Name
-a----	9/24/2026 8:45 PM	62	jsmith-test.txt

```
PS C:\WINDOWS\system32>
```

At the bottom right of the window, a message states: "Windows License is expired Build 26100.ge_release.240331-1435". The taskbar at the bottom shows the time as 10:57 PM on 9/24/2026.

Project Outcome & Skills Demonstrated

The completed lab demonstrates Windows Server administration, Active Directory identity and group management, SMB file sharing, NTFS permissions, Group Policy Preferences, Windows 11 domain-client administration, PowerShell verification, and structured troubleshooting.

Next phase: Deploy separate IT, HR and Sales shares with dedicated security groups, least-privilege NTFS permissions, mapped drives and cross-department access testing.

LinkedIn caption: Hands-On Windows Server 2025 Project: Active Directory, File Server, NTFS Permissions & Group Policy